

Venta

Technical Specification Document

E-Commerce Platform · v1.0

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DATE: August 8, 2026

STATUS: Draft

1. INTRODUCTION

1.1 Project Overview

Venta is a web application that connects buyers and sellers of used computer components. Customers can browse listings and communicate with sellers within the platform. Moderators review all listings before being published on the site and users will be notified once their listing has been approved.

1.2 Purpose of This Document

This Technical Specification Document (TSD) defines the architecture, features, data models, and API design for Venta v1.0. It serves as the primary reference for the developer throughout the software development lifecycle.

1.3 Scope

This document covers the following components of the platform:

- Customer-facing web application
- Admin management dashboard
- Backend REST API
- Database schema and data models

1.4 Definitions

TERM	DEFINITION
Buyer	A registered user who can browse listings and transact with sellers.
Seller	A registered user who can create and manage his/her listings on the platform.
Moderator	An admin user who can approve or take down listings and suspend users.
Listing	A product posted by a seller for sale on the platform.

2. TECHNOLOGY STACK

2.1 Frontend

LAYER	TECHNOLOGY	REASON
Web Application	React 19	Better scalability and performance through SPA
Styling	Custom CSS, Tailwind	Consistent design and faster implementation
State Management	zustand	Lightweight, comes with React

2.2 Backend

LAYER	TECHNOLOGY	REASON
Runtime	Node.js	Large ecosystem
Framework	Express.js	Minimal, flexible REST API framework
Authentication	JWT	Solid, scalable authentication

2.3 Database

DATABASE	TECHNOLOGY	USAGE
Primary DB	MongoDB	Users, Listings, Messages

3. SYSTEM ARCHITECTURE

3.1 Platform Architecture

The frontend client communicates with the backend REST API over Axios. The API handles business logic and interfaces with the MongoDB database.

3.2 Architecture Layers

LAYER	COMPONENTS	RESPONSIBILITY
Presentation	React 19 + Tailwind	UI rendering, user experience, state management
API	Express.js REST API	Request routing, authentication, validation
Data	MongoDB	Store, query, and update rapidly changing, structured application data.
Storage	Amazon S3	Store and serve static, unstructured files

3.3 Deployment

- Frontend deployed on Amazon S3
- Backend API hosted on Render

4. FEATURE SPECIFICATIONS

4.1 Buyer Features

FEATURE	DESCRIPTION	PRIORITY
Registration and Login	Sign up with email and username	High
Browse Listings	View all approved listings	High
Chat with Seller	Chat with the seller for a possible transaction	Medium
Search Filters	Filter listings by category and date	Medium

4.2 Seller Features

FEATURE	DESCRIPTION	PRIORITY
Add Listings	Add listings with photos and accurate item information for moderator approval	High
Chat with Buyer	Chat with potential buyers	Medium
Update Listing Status	Update listing status to Available or Sold	High

4.3 Admin Features

FEATURE	DESCRIPTION	PRIORITY
Approve / Deny Listing	Approve or deny seller listings with a reason	High
Suspend Buyers or Sellers	Block users due to violation of Terms of Service	Medium
Dashboard	Monitor published listings, pending listings, ratings, and reports	High

5. DATABASE SCHEMA

5.1 Models

DOCUMENT	PROPERTIES	DESCRIPTION
User	id, username, email, password, phone, role, status	Platform users (buyers and sellers)
Listing	id, userId, price, category, title, description, location, photos, status, dealMethod	Seller listings including proof of item
Message	id, conversationId, senderId, content	Individual messages between buyer and seller
Conversation	id, buyerId, sellerId, listingId, lastMessage, lastMessageAt	Chat thread linked to a specific listing

5.2 Listing Status Flow

A listing transitions through the following statuses:

- Pending - listing submitted by seller, awaiting moderator review
- Available - listing approved by moderator and visible to buyers
- Sold - listing marked as sold by the seller after a completed transaction
- Suspended - listing taken down by a moderator due to a violation

6. API DESIGN

6.1 Authentication Endpoints

METHOD	ENDPOINT	DESCRIPTION	AUTH
POST	/api/register	Register new user	None
POST	/api/login	Login and generate JWT	None
GET	/api/:username	Get profile details	Token
POST	/api/logout	Logout, clear token	Token

6.2 Listing Endpoints

METHOD	ENDPOINT	DESCRIPTION	AUTH
POST	/api/listings	Create a new listing	Token
GET	/api/listings	View all available listings	None
GET	/api/listings/:listingId	View a single listing	None
PATCH	/api/listings/:listingId	Update listing details or status	Token
DELETE	/api/listings/:listingId	Delete listing	Token / Admin
GET	/api/dashboard	View all published listings	Admin
GET	/api/dashboard/pending	View all pending listings	Admin
PATCH	/api/dashboard/publish/:listingId	Approve pending listing	Admin
PATCH	/api/dashboard/suspended/:listingId	Suspend listing	Admin

6.3 Messaging Endpoints

METHOD	ENDPOINT	DESCRIPTION	AUTH
POST	/api/conversations	Start a conversation about a listing	Token
GET	/api/conversations	Get all conversations for a user	Token
GET	/api/conversations/:conversationId	Get messages in a conversation	Token
POST	/api/conversations/:conversationId/messages	Send a message	Token

7. SECURITY CONSIDERATIONS

7.1 Authentication and Authorization

- JWT access tokens expire after 1 day
- Role-based access control enforced at the API middleware level
- All passwords hashed using bcrypt with a salt of 12

7.2 Data Protection

- Database access restricted to application server IP only
- Environment variables are used for all secrets and are never hardcoded
- All data in transit encrypted via HTTPS

7.3 API Security

- CORS configured to allow only approved frontend origins
- Input validation applied on all endpoints to prevent malformed requests

8. NON-FUNCTIONAL REQUIREMENTS

REQUIREMENT	TARGET	NOTES
API Response Time	< 500ms	For standard listing and user requests
Scalability	100+	The architecture should allow the application to scale as user traffic increases.

9. RISKS AND MITIGATIONS

RISK	LIKELIHOOD	IMPACT	MITIGATION
Data Breach	Low	Critical	Encryption, regular security audits, principle of least privilege
Reputational Risks	Medium	High	Moderator review of all listings and, ability to suspend accounts
Fake or Fraudulent Listings	Medium	High	Require photos and accurate pricing before approval, moderator verification
Scope Creep	High	Medium	Strict milestone reviews, any new features deferred to v2.0

10. DEVELOPMENT MILESTONES

MILESTONE	SCOPE	TARGET
v0.1 – Planning & Design	TSD, DB schema, wireframes, API design	Days 1 to 4
v0.2 – Auth & Profiles	Registration, login, user profiles	Week 1 to 2
v0.3 – Listings	Create listings, proof upload, status management	Week 2 to 3
v0.4 – Browse & Search	Browse, search, filters	Week 3 to 4
v0.5 – Messaging	In-app conversation between buyer and seller	Week 4 to 5
v0.6 – Admin Panel	Moderation dashboard, approve/reject/suspend	Week 6 to 7
v0.7 – Static Pages & Polish	Terms, Privacy Policy, FAQ, UI cleanup	Week 7
v1.0 – Launch	Testing, bug fixes, deployment	Week 8